

Muhammad Azeem Bhatti

AI/ML Engineer

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AI/ML engineer focused on explainable, medical-imaging machine learning — from feature selection to deployed inference pipelines.

EDUCATION

MS, Artificial Intelligence

Air University, E-9 Campus, Islamabad

Sep 2025 – Present · CGPA 3.97/4.0 · Advisor: Dr. Mehdi

BS, Electrical Engineering (Electronics)

Air University, E-9 Campus, Islamabad

Sep 2015 – June 2019

EXPERIENCE

AI Intern — GoldenGateInnovations

Jul 2026 – Present

Speech Transcription & Translation Pipeline (Urdu–English)

- Built a production speech transcription/translation pipeline for Urdu-English code-switched audio (faster-whisper large-v3 ASR + Fireworks GLM-5.2 LLM translation), cutting translation WER from ~62% to ~46% via a custom translate-critique-refine pipeline with tuned prompts and retry/batching logic.
- Validated the system against a human-annotated Urdu-English benchmark, achieving 28.9% WER as an externally-verifiable accuracy metric.
- Added real-time speaker diarization (pyannote.audio) to the GPU inference pipeline, resolving production issues spanning cloud model-access restrictions, cross-GPU CUDA incompatibility, and NumPy/PyTorch ABI conflicts.
- Built a bilingual PDF transcript export handling right-to-left text shaping and script detection, and deployed/maintained the GPU inference service on RunPod with Docker and zero-downtime rollouts.

OSINT News Intelligence Pipeline (Pakistan-Interest News Monitoring System)

Python, SQLite, SQLAlchemy, Streamlit, Groq LLM API, RSS/PRAW, APScheduler

- Built an end-to-end OSINT pipeline in Python collecting, deduplicating, and classifying news from 8+ live sources (RSS, Reddit, government portals) into a structured SQLite database.
- Integrated an LLM classifier (Groq API) to tag articles by genre and national-interest alignment, producing structured JSON output with confidence scores across 500+ articles.
- Designed a pluggable source-connector architecture for adding new sources without touching the core pipeline, with content-hash deduplication and APScheduler-driven scheduled batch processing.

K-Electric Automated Meter Reading

- Built an automated meter-reading system using rotated YOLO bounding boxes and OCR to localize and read K-Electric utility meters from field photos, achieving 94% exact-reading accuracy on a 133-photo test set.

PROJECTS

Effectiveness of Balancing Plant's Dataset using GAN

Semester Project

Used a DCGAN to synthetically balance a 38-class crop leaf disease image dataset, generating additional samples for underrepresented classes ahead of downstream classification; the balanced dataset reached 98.19% accuracy and 0.982 macro-F1.

Privacy-Preserving Techniques in Federated Learning

Semester Project

Built a PyTorch-based federated learning pipeline combining Hensel's compression, differential privacy on model weights, and secure aggregation to protect client data throughout distributed training.

LatentGAT

Open-Source Project · Sep 2026

Built a 3D U-Net with graph attention for brain tumor segmentation, reaching a Dice score of 0.869 when evaluated on BraTS2021 and two unseen external cohorts.

PUBLICATIONS

Multi-Objective Selection of Hybrid Deep–Radiomic Features for Brain Tumor MRI Classification with Transparent Fuzzy Rule-Based Inference

Information Fusion (Elsevier) · First Author · Submitted, with editor

Proposed OHHO-NSGA-II — a three-objective, Opposition-Based Harris Hawks-driven extension of NSGA-II — paired with an ANFIS neuro-symbolic classifier for inherently interpretable brain tumor MRI classification. Reduced a 2,420-feature hybrid pool to 25 (98.97% search-space reduction), reaching 94.45% cross-validation accuracy and 0.9383 macro-F1 on the Figshare MRI dataset.

A Privacy-Preserving Federated Learning Framework with Variance-Based Aggregation and Customized U-Net for Brain Tumor Segmentation

Frontiers in Oncology · Third Author · Ready to submit

Designed Fed-PLM, a federated learning framework pairing a lightweight attention-based U-Net with a novel aggregation method that identifies the dominant latent principal mode across client updates, reaching a whole-tumor Dice of 0.885 under federation versus 0.894 centralized on BraTS 2021.

SKILLS

LANGUAGES

Python

ML/DL & RESEARCH

PyTorch GANs (DCGAN) U-Net

Graph Attention Networks ANFIS

Federated Learning NSGA-II

Harris Hawks Opt. Explainable AI

Radiomics YOLO CRNN OCR Grad-CAM

NLP & SPEECH AI

LLM Integration faster-whisper (ASR)

Speaker Diarization

Code-Switching Detection

Machine Translation

TOOLS

PyCharm VS Code GitHub